

9.4.2

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Question: Find the roots of the following quadratic equation graphically:

$$x(2x + 3) = x^2 + 1 \quad (0.1)$$

Solution: First, we expand and rearrange the equation to standard form:

$$x(2x + 3) = x^2 + 1 \quad (0.2)$$

$$2x^2 + 3x = x^2 + 1 \quad (0.3)$$

$$2x^2 + 3x - x^2 - 1 = 0 \quad (0.4)$$

$$x^2 + 3x - 1 = 0 \quad (0.5)$$

Input Variables:

The given quadratic can be written in the conic form:

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (0.6)$$

where

$$\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix}, \quad f = -1 \quad (0.7)$$

Since the roots of the quadratic correspond to intersections with the x-axis, we represent the line:

$$L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m} \quad (0.8)$$

with

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (0.9)$$

Symbol	Value
$\mathbf{V}$	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
$\mathbf{u}$	$\begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix}$
f	-1
$\mathbf{h}$	$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$
$\mathbf{m}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

TABLE 0: Input Parameters

The points of intersection of a line with a conic are given by:

$$\kappa = \frac{1}{\mathbf{m}^T \mathbf{V} \mathbf{m}} \left[-\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{(\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u}))^2 - g(\mathbf{h}) (\mathbf{m}^T \mathbf{V} \mathbf{m})} \right] \quad (0.10)$$

where

$$g(\mathbf{h}) = \mathbf{h}^T \mathbf{V} \mathbf{h} + 2\mathbf{u}^T \mathbf{h} + f \quad (0.11)$$

Step 1: Compute $\mathbf{m}^T \mathbf{V} \mathbf{m}$

$$\mathbf{m}^T \mathbf{V} \mathbf{m} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \quad (0.12)$$

Step 2: Compute $\mathbf{V} \mathbf{h} + \mathbf{u}$

$$\mathbf{V} \mathbf{h} + \mathbf{u} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} \quad (0.13)$$

Step 3: Compute $\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u})$

$$\mathbf{m}^T (\mathbf{V} \mathbf{h} + \mathbf{u}) = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} = \frac{3}{2} \quad (0.14)$$

Step 4: Compute $g(\mathbf{h})$

$$g(\mathbf{h}) = \mathbf{h}^T \mathbf{V} \mathbf{h} + 2\mathbf{u}^T \mathbf{h} + f = 0 + 0 + (-1) = -1 \quad (0.15)$$

Step 5: Substitute into formula for κ

$$\kappa = -\frac{3}{2} \pm \sqrt{\left(\frac{3}{2}\right)^2 - (-1)(1)} \quad (0.16)$$

$$= -\frac{3}{2} \pm \sqrt{\frac{9}{4} + 1} \quad (0.17)$$

$$= -\frac{3}{2} \pm \sqrt{\frac{13}{4}} \quad (0.18)$$

$$= -\frac{3}{2} \pm \frac{\sqrt{13}}{2} \quad (0.19)$$

$$= \frac{-3 \pm \sqrt{13}}{2} \quad (0.20)$$

Step 6: Evaluate roots

$$\kappa_1 = \frac{-3 + \sqrt{13}}{2} \quad (0.21)$$

$$\kappa_2 = \frac{-3 - \sqrt{13}}{2} \quad (0.22)$$

Step 7: Find intersection points The intersection points are obtained as:

$$\mathbf{x}_1 = \mathbf{h} + \kappa_1 \mathbf{m} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \frac{-3 + \sqrt{13}}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{-3 + \sqrt{13}}{2} \\ 0 \end{pmatrix} \quad (0.23)$$

$$\mathbf{x}_2 = \mathbf{h} + \kappa_2 \mathbf{m} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \frac{-3 - \sqrt{13}}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{-3 - \sqrt{13}}{2} \\ 0 \end{pmatrix} \quad (0.24)$$

Thus, the quadratic $x^2 + 3x - 1 = 0$ intersects the x-axis at:

$$x = \frac{-3 - \sqrt{13}}{2} \approx -3.303 \quad (0.25)$$

$$x = \frac{-3 + \sqrt{13}}{2} \approx 0.303 \quad (0.26)$$

The roots of the original equation $x(2x + 3) = x^2 + 1$ are:

$$\boxed{x = \frac{-3 \pm \sqrt{13}}{2}} \quad (0.27)$$

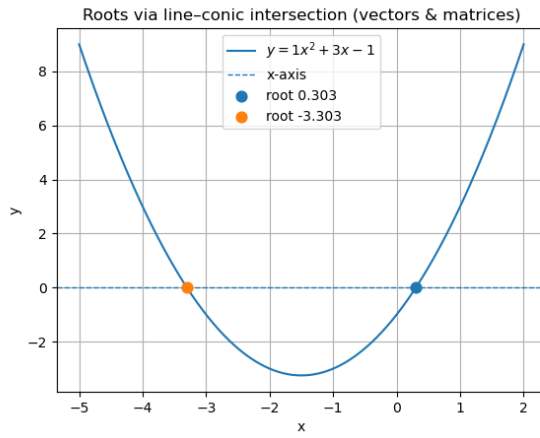


Fig. 0.1: Vector Representation